



Modelling Financial Markets

Chapter 10: Market Microstructure



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Acronyms used in this material

- AMM: Automated Market Maker
- BET: Bucharest Exchange Trading (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CEX: Centralised EXchange
- CFTC: Commodity Futures Trading Commission
- DEX: Decentralised EXchange
- ETF: Exchange-Traded Fund
- FOK: Fill-Or-Kill (order)
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- GTC: Good-Till-Cancelled (order)
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HFT: High-Frequency Trading
- IOC: Immediate-Or-Cancel (order)
- LOB: Limit Order Book
- NYSE: New York Stock Exchange
- PIN: Probability of INformed trading
- SEC: Securities and Exchange Commission
- TWAP: Time-Weighted Average Price
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)
- VIX: CBOE Volatility Index
- VPIN: Volume-synchronised Probability of INformed trading
- VWAP: Volume-Weighted Average Price
- XRP: XRP (the native token of the XRP Ledger)

Outline

Motivation

The Limit Order Book

The Bid-Ask Spread

Illiquidity and Price Impact

Intraday Patterns

Information and Prices

Price Impact and Optimal Execution

High-Frequency Trading and Market Stress

Crypto Market Microstructure

Liquidity at the Bucharest Stock Exchange

Summary

References

What Is Market Microstructure?

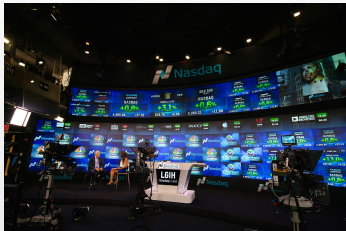
The question of this chapter

- Earlier chapters took prices as given: a daily close, a return, a volatility
- Here we ask **how** a price is formed: who trades, with which orders, under which rules
- And what trading **costs**: the spread, the price impact, the delay [Harris \(2003\)](#)

What we do in this chapter

- Describe the limit order book and simulate one
- Measure spreads and illiquidity for US stocks, BVB stocks and crypto-assets
- Explain spreads by adverse selection (Glosten–Milgrom, Kyle)
- Model price impact and optimal execution; study high-frequency trading and market stress

From the Trading Floor to the Matching Engine



Nasdaq MarketSite, Times Square, New York: a studio,
not a trading floor

Photo: Luca Marfè (2015); CC BY 2.0; Wikimedia Commons

- For two centuries prices were made by people in a room: brokers, specialists, open outcry
 - ▶ The specialist kept the order book on paper and quoted a bid and an ask
- Today almost all equity, futures and crypto trading runs on computers that match orders in microseconds
- The rules of the matching engine (priority, tick size, auctions) shape prices and costs [Gould et al. \(2013\)](#)
- Textbooks: [Hasbrouck \(2007\)](#), [Bouchaud, Bonart, Donier & Gould \(2018\)](#), [Franke, Härdle & Hafner \(2019\)](#)

Data Used in This Chapter

Series	Content	Period
SPY, 5-minute bars	ETF on the S&P 500; regular session 09:30–16:00 New York time; 1 477 full days	2020–2026
Bitcoin, 5-minute bars	24 hours a day, 7 days a week; 208 596 bars	2024–2026
US stocks, daily	SPY, Apple, Microsoft, Nvidia, JPMorgan, GameStop, Strategy	2016–2026
BVB stocks, daily	Ten blue chips and the BET ETF; values in lei converted to USD at the BNR reference rate	2016–2026
Crypto-assets, daily	Bitcoin, Ethereum, Solana, XRP, Dogecoin, Cardano, Litecoin	2016–2026
VIX, daily	Implied volatility of the S&P 500	2020–2026

- Daily data: open, high, low, close and volume; prices adjusted for dividends and splits; days without trades removed
- Order books in Section 2 and the bid-ask bounce in Section 3 are **simulated**, and labelled as such

Liquidity: Four Dimensions

- **Liquidity**: the ability to trade a quantity quickly, at a price close to the fair value
- Three dimensions named by [Kyle \(1985\)](#)
 - ▶ **Tightness**: the cost of turning a position around quickly (the bid-ask spread)
 - ▶ **Depth**: the order size needed to move the price by a given amount
 - ▶ **Resiliency**: how fast the price returns after an uninformed shock
- **Immediacy**: how long one must wait for execution [Harris \(2003\)](#)
- No single number captures all four: this chapter uses several measures and compares them

Order Types

- **Market order:** buy or sell a quantity now, at the best available prices; certain execution, uncertain price
- **Limit order:** buy at most at price p (or sell at least at p); certain price, uncertain execution
- Time conditions
 - ▶ IOC (Immediate-Or-Cancel): execute what is possible now, cancel the rest; FOK (Fill-Or-Kill): all now or nothing
 - ▶ GTC (Good-Till-Cancelled): stays in the book until executed or withdrawn
- **Stop order:** becomes a market order when the price crosses a trigger; **iceberg:** shows only part of its size
- Limit orders **supply** liquidity; market orders **consume** it

The Limit Order Book: Definitions

- **Limit order book** (LOB): the list of all standing limit orders, by price and time of arrival [Gould et al. \(2013\)](#)
- Best bid b_t (highest buy price) and best ask a_t (lowest sell price), always $b_t < a_t$
 - ▶ Mid-price $m_t = (a_t + b_t)/2$; quoted spread $a_t - b_t$; relative spread $(a_t - b_t)/m_t$, usually in basis points (1 bp = 0.01%)
- **Depth** at a price level: the total quantity waiting there
- **Tick size**: the smallest allowed price step (USD 0.01 for most US stocks priced above USD 1)
- A trade happens only when an incoming order crosses the opposite best quote

Price-Time Priority and the Matching Engine

- **Price priority:** a better price is served first
 - ▶ For buyers the highest bid, for sellers the lowest ask
- **Time priority:** at the same price, the earlier order is served first (a queue)
- Being first in the queue is valuable: a reason to invest in speed (Section 8)
- Many exchanges open and close with a **call auction**: orders are collected and matched at a single price
 - ▶ The single price maximises the executed quantity

Worked Example: A Market Order Walks the Book

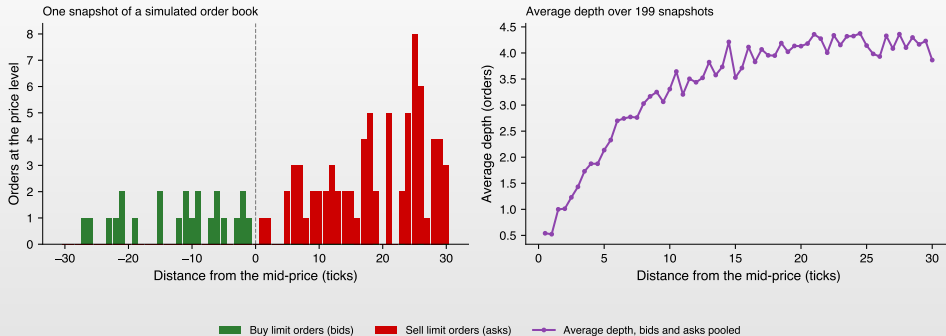
Bids		Asks	
400	99.98	100.02	300
600	99.97	100.03	500
800	99.95	100.05	400
1 500	99.90	100.08	1 000

- Mid-price $m = 100.00$; quoted spread 0.04, i.e. 4 bp
- Market buy of 800 shares: 300 at 100.02, then 500 at 100.03
 - ▶ Average price $(300 \cdot 100.02 + 500 \cdot 100.03)/800 = 100.02625$
- Cost relative to the mid-price: 2.63 bp; relative to the best ask: 0.63 bp
- After the trade the best ask is 100.05: the spread widened to 0.07

A Simple Model of the Order Book

- **Zero-intelligence** model: orders arrive at random, with no strategy [Farmer, Patelli & Zovko \(2005\)](#)
 - ▶ Limit orders: rate α per price level, placed uniformly up to 40 ticks from the opposite best quote
 - ▶ Market orders: rate μ per side; cancellations: rate δ per standing order
- Here $\alpha = 0.1$, $\mu = 0.5$, $\delta = 0.02$; 300 000 events, unit orders
- Farmer, Patelli and Zovko show that such a model already predicts much of the cross-section of spreads on the London Stock Exchange
- Use: a laboratory where spread, depth and impact can be measured exactly

A Simulated Order Book

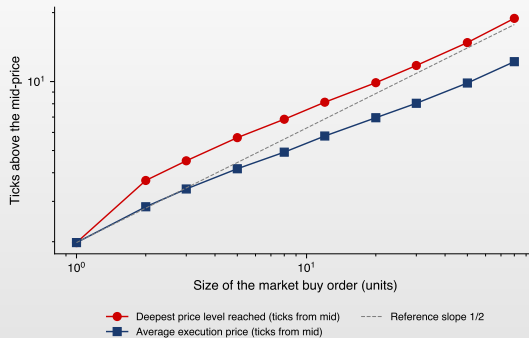


□ Left: one snapshot (bids green, asks red); right: average depth by distance from the mid, bids and asks pooled

Reading the Simulated Order Book

- Mean spread 3.86 ticks, median 3; in 10% of the time it exceeds 7 ticks
- Depth is thin near the mid and grows away from it
 - ▶ 0.7 orders at 1 tick, 3.3 at 10 ticks, 4.2 at 25 ticks (average over 199 snapshots)
- Reason: market orders eat the best levels first, so liquidity near the price is scarce [Tóth et al. \(2011\)](#)
- The same hump shape is observed in real order books [Bouchaud, Bonart, Donier & Gould \(2018\)](#)

The Price Impact of a Market Order



Market buy orders of 1 to 80 units sent against each simulated snapshot; log-log axes

Reading the Impact Function

- ▣ A 1-unit order reaches 2.0 ticks above the mid; an 80-unit order reaches 18.9 ticks
- ▣ **Concave** impact: slope 0.47 on log-log axes, close to $1/2$
 - ▶ Average execution price: slope 0.40; 80 units cost 12.2 ticks on average
- ▣ Doubling the order does not double the cost: depth grows with distance from the mid
- ▣ Real markets show a similar concave law for large orders split over time (Section 7)

Recap: The Order Book

- Limit orders supply liquidity, market orders consume it; price and time decide who trades first
- A market order pays half the spread plus the cost of walking into deeper levels
- Even random order flow creates a spread, a hump-shaped depth profile and concave impact [Farmer, Patelli & Zovko \(2005\)](#)

Why Is There a Spread?

- ▣ A liquidity provider buys at the bid and sells at the ask; the spread pays three costs [Huang & Stoll \(1997\)](#)
- ▣ **Order processing:** fees, technology, clearing
- ▣ **Inventory:** holding an unwanted position exposes the provider to price risk
- ▣ **Adverse selection:** some counterparties know more [Bagehot \(1971\)](#)
 - ▶ The provider loses to informed traders and must recover it from uninformed ones (Section 6)

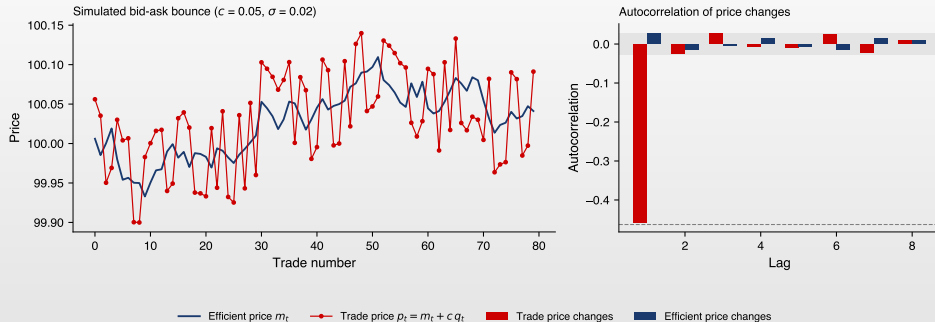
Quoted, Effective and Realised Spread

- ▣ Trade at price p_t , direction $q_t = +1$ (buyer-initiated) or -1 (seller-initiated), mid-price m_t
- ▣ **Quoted:** $S_t^Q = (a_t - b_t)/m_t$; **effective:** $S_t^E = 2q_t(p_t - m_t)/m_t$, the cost actually paid
- ▣ **Realised:** $S_t^R = 2q_t(p_t - m_{t+\Delta})/m_t$, the provider's gain after Δ (e.g. 5 minutes)
 - ▶ Price impact = $S_t^E - S_t^R = 2q_t(m_{t+\Delta} - m_t)/m_t$: the part lost to informed flow
- ▣ All three need trades, quotes and the trade direction, signed for example by the rule of [Lee & Ready \(1991\)](#)
- ▣ Without such data we **estimate** the spread from prices only: the next slides

The Roll Model

- Efficient price: random walk $m_t = m_{t-1} + \varepsilon_t$, $\varepsilon_t \sim (0, \sigma^2)$ i.i.d. [Roll \(1984\)](#)
 - ▶ Trade price $p_t = m_t + c q_t$, with half-spread c and $q_t = \pm 1$ with probability $1/2$, independent of ε
- Price change: $\Delta p_t = \varepsilon_t + c(q_t - q_{t-1})$
- $\text{Var}(\Delta p_t) = \sigma^2 + 2c^2$, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$
- **Roll estimator:** $\hat{s} = 2\hat{c} = 2\sqrt{-\widehat{\text{Cov}}(\Delta p_t, \Delta p_{t-1})}$
 - ▶ The bid-ask bounce creates negative autocorrelation even when the efficient price is a random walk

The Bid-Ask Bounce (Simulated)



- 5 000 simulated trades with $c = 0.05$ and $\sigma = 0.02$; grey band: $\pm 1.96/\sqrt{n}$; dashed line: the theoretical first-order autocorrelation

Reading the Roll Simulation

- True spread $2c = 0.10$; Roll estimate from the simulated trades: $\hat{s} = 0.0998$
- First-order autocorrelation -0.458 , theory $-c^2/(\sigma^2 + 2c^2) = -0.463$
- No autocorrelation at lags 2 and beyond: the bounce lasts only one trade
- The efficient price changes are uncorrelated: the market is efficient, yet trade prices are not a random walk

When the Roll Estimator Fails

- With real data the sample covariance is often **positive**: $\hat{s}$ does not exist
 - ▶ SPY, 5-minute bars: the covariance within the day is positive on 41% of days
- Causes: momentum or news trends, orders split into many trades, stale prices, bars instead of trades
- With daily data the bounce (a few bp) is buried under the daily volatility (103 bp for SPY over the last two years)
- Hence estimators based on the daily high and low: the next two slides

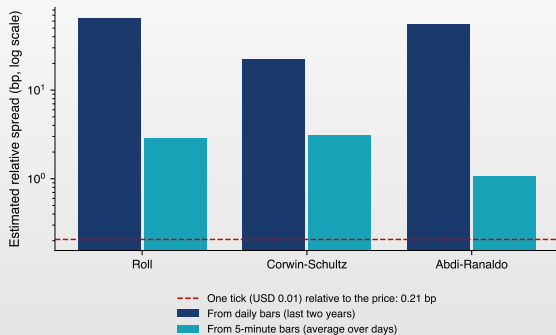
The Corwin–Schultz Estimator

- Idea: the daily high is usually a buy at the ask, the low a sell at the bid; the high-low range = volatility + spread [Corwin & Schultz \(2012\)](#)
- The volatility part grows with the length of the interval, the spread part does not: compare one day with two days
- $\beta = \sum_{j=0}^1 \left[\ln \frac{H_{t+j}}{L_{t+j}} \right]^2$, $\gamma = \left[\ln \frac{H_{t,t+1}}{L_{t,t+1}} \right]^2$, $\alpha = \frac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma}{3 - 2\sqrt{2}}}$
- $\hat{S} = \frac{2(e^\alpha - 1)}{1 + e^\alpha}$, with $H_{t,t+1}$, $L_{t,t+1}$ the high and low over the two days
 - Negative two-day estimates are set to 0, then averaged over the period

The Abdi–Ranaldo Estimator

- ▣ Uses the close c_t and the mid-range $\eta_t = (\ln H_t + \ln L_t)/2$ as a proxy for the mid-price [Abdi & Ranaldo \(2017\)](#)
- ▣ $\hat{S}^2 = 4 \mathbb{E}[(c_t - \eta_t)(c_t - \eta_{t+1})]$, with c_t the log close
- ▣ The close sits at the bid or the ask; the distance to the neighbouring mid-ranges reveals the half-spread
- ▣ If the sample mean is negative, the estimate is set to 0
- ▣ For crypto-assets, both high-low estimators track the time variation of high-frequency liquidity well [Brauneis, Mestel, Riordan & Theissen \(2021\)](#)
 - ▶ The same study finds the Amihud ratio better for **levels** and for comparing venues

Spread Estimators at Two Frequencies: SPY



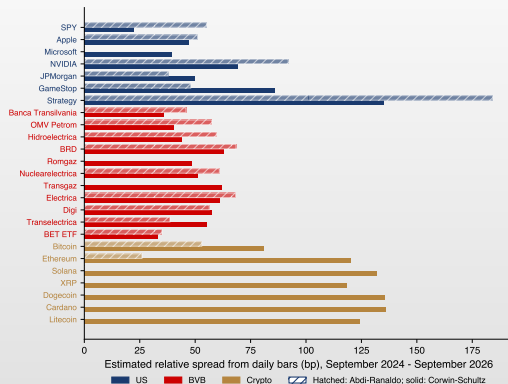
□ Daily bars: September 2024 – September 2026; 5-minute bars: estimator per day, averaged over 1 477 days (2020–2026)

 MFM_ch10_spread_estimators

Reading the Frequency Chart

- A spread cannot be smaller than one price step; with a step of USD 0.01 the bound for SPY is 0.21 bp
- From daily bars: Roll 64.3 bp, Corwin–Schultz 22.4 bp, Abdi–Ranaldo 55.1 bp
 - ▶ Two orders of magnitude above the tick: the estimators measure volatility, not the spread
- From 5-minute bars: 2.82, 3.11 and 1.06 bp: much closer, still above the tick bound
- Lesson: for very liquid assets, low-frequency spread proxies are dominated by volatility

Spread Estimates Across 25 Assets



□ Daily bars, September 2024 – September 2026; BVB stocks in lei, prices adjusted for dividends and splits

Reading the Cross-Section of Spread Estimates

- Median Corwin–Schultz estimate: US 50 bp, BVB 51 bp, crypto 124 bp
 - ▶ BVB blue chips look as liquid as US large caps: implausible, since they trade a thousand times less value (next section)
- The ranking follows volatility: crypto-assets and the most volatile stocks get the widest estimates
- Abdi–Rinaldo is 0 for 8 of the 25 assets: a negative sample mean
- Use these estimators for changes over time within one asset, not for levels across assets [Brauneis, Mestel, Riordan & Theissen \(2021\)](#)

Recap: Spreads

- The spread pays for order processing, inventory risk and adverse selection
- Bid-ask bounce makes trade prices negatively autocorrelated: the Roll estimator
- High-low estimators work at high frequency; on daily data they mostly measure volatility

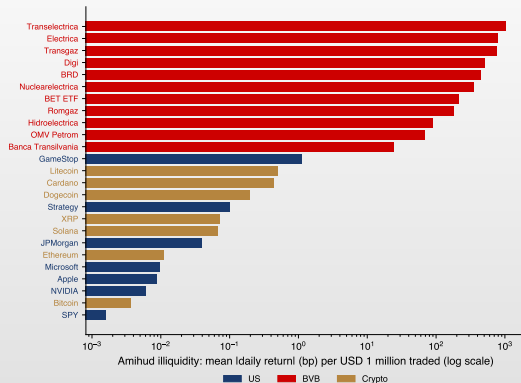
The Amihud Illiquidity Ratio

- Daily price impact per unit of money traded [Amihud \(2002\)](#)

►
$$\text{ILLIQ}_i = \frac{1}{D_i} \sum_{d=1}^{D_i} \frac{|r_{i,d}|}{\text{DVOL}_{i,d}}, \text{ with } r_{i,d} \text{ the daily return and } \text{DVOL}_{i,d} \text{ the traded value}$$

- Units here: basis points of price move per USD 1 million traded
- Needs only daily returns and volumes: available for every market and every period
- A low-frequency proxy of the Kyle price-impact coefficient (Section 6) [Hasbrouck \(2009\)](#)
- Traded value of BVB stocks: value in lei divided by the BNR reference rate of the day; crypto: traded value in USD summed across exchanges

Amihud Illiquidity Across 25 Assets

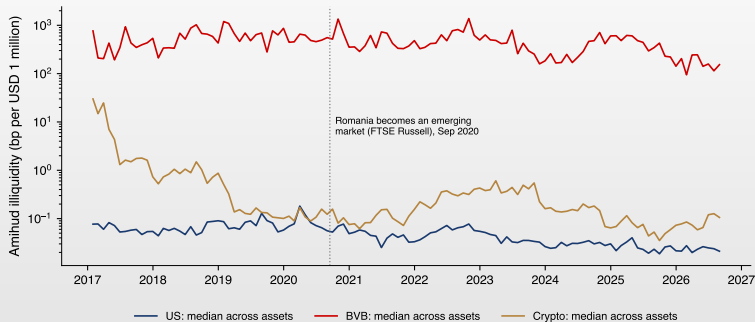


September 2024 – September 2026; log scale; each bar is the mean over about 500 trading days (730 for crypto)

Reading the Amihud Ratios

- ▣ SPY: 0.0016 bp per USD 1 million; median daily value USD 40 124 million
 - ▶ Bitcoin: 0.0037 bp, with USD 41 834 million a day across exchanges
- ▣ Banca Transilvania, the most liquid BVB stock: 24.7 bp; median value USD 3.90 million a day
 - ▶ About 15 541 times the SPY figure; Transelectrica: 1024.0 bp
- ▣ Three clusters: US large caps and the top crypto-assets, then smaller crypto-assets and GameStop, then the BVB
- ▣ Unlike the spread estimators, the Amihud ratio separates markets by orders of magnitude

Illiquidity over Time: Three Markets

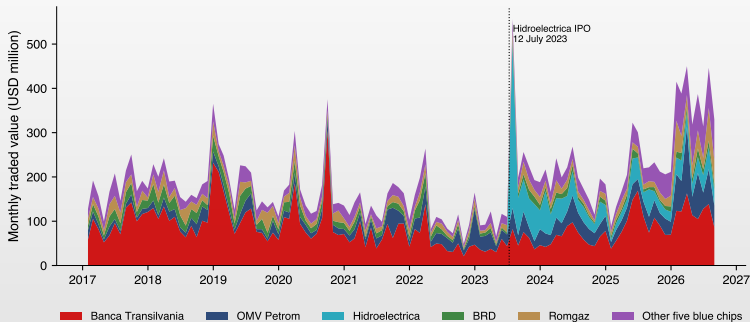


□ Monthly Amihud ratio, median across the assets of each group (without SPY and the BET ETF); log scale

Reading Illiquidity over Time

- ▣ Crypto: from 7.6 bp per USD 1 million in 2017 to 0.089 in 2026
 - ▶ Crypto trading grew from a niche to institutional volumes; spot ETFs followed (Chapter 16)
- ▣ BVB: from 435 to 169: more liquid, but still about 7 212 times the US median
- ▣ US: from 0.062 to 0.023: large caps became even cheaper to trade
- ▣ Romania moved from frontier to emerging-market status in September 2020 [BVB \(history\)](#); the BVB median shows no break at that date

Traded Value of BVB Blue Chips



□ Monthly sum of daily traded value, ten blue chips, converted to USD at the BNR reference rate

Reading the BVB Traded Value

- ▣ Ten blue chips: USD 2 095 million traded in 2017, 1 494 million in 2022, 2 520 million in 2025
- ▣ Concentration: Banca Transilvania alone made 44% of the 2025 value, Hidroelectrica 13%
 - ▶ The Hidroelectrica listing of July 2023 added a second large, liquid stock
- ▣ One day of SPY trading exceeds a whole year of the ten BVB blue chips
- ▣ Consequence for a fund: a position of a few million USD takes days to build or unwind

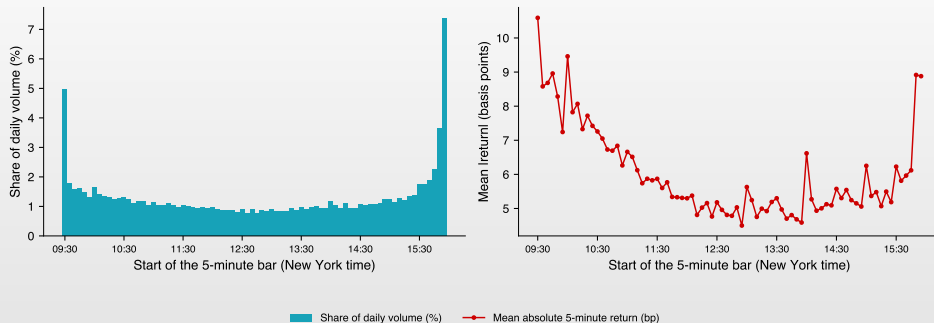
Is Liquidity Priced?

- **Level:** investors demand a higher return for assets that are costly to trade [Amihud & Mendelson \(1986\)](#)
 - ▶ Expected return increases with the spread, and concavely: long-horizon investors hold the illiquid assets
- **Time series:** expected market illiquidity raises future returns; an unexpected rise lowers current prices [Amihud \(2002\)](#)
- **Liquidity risk:** stocks that fall when market liquidity dries up earn a premium [Pástor & Stambaugh \(2003\)](#), [Acharya & Pedersen \(2005\)](#)
- Seminar Part C tests the time-series prediction on the BVB, US and crypto markets

Recap: Illiquidity

- The Amihud ratio measures price impact per unit of value traded; it separates markets by orders of magnitude
- BVB blue chips need thousands of times more price move per dollar than US large caps
- Liquidity has a level premium and a risk premium; illiquidity shocks lower prices

SPY Within the Trading Day



- 78 five-minute bars per day, 1 477 full days (October 2020 – September 2026); the first bar's return starts at the opening price

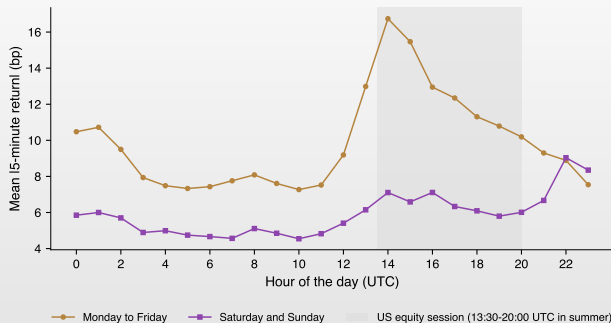
Reading the U-Shape

- ▣ Volume: the first bar carries 5.0% of the day, the last 7.4%; the last 30 minutes 18.6%
 - ▶ A midday bar (11:30–14:00) carries only 0.9% on average
- ▣ Volatility: mean absolute return 10.6 bp in the first bar, 5.1 bp at midday, 8.9 bp at the close
- ▣ The opening bar moves 2.06 times as much as a midday bar
- ▣ Spikes at 10:00 and 14:00: scheduled macro releases and Federal Reserve statements

Why a U-Shape?

- Documented for NYSE stocks by [Wood, McInish & Ord \(1985\)](#): returns and volume are highest at the open and the close
- **Open**: overnight news is absorbed; uncertainty is highest
 - ▶ Informed traders act early, before the information spreads
- **Clustering**: uninformed traders who can choose the time to trade together, and liquidity attracts liquidity [Admati & Pfleiderer \(1988\)](#)
- **Close**: index funds and ETFs trade at the closing price, which is fixed by the closing auction
- Practical rule: an execution algorithm follows the volume curve to reduce impact (Section 7)

Bitcoin Around the Clock

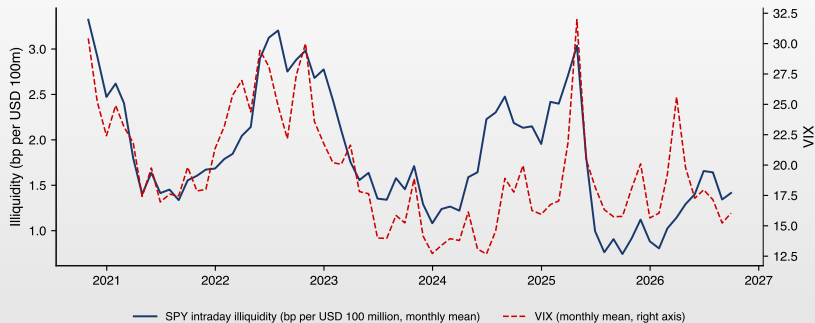


□ Mean absolute 5-minute return by hour (UTC), September 2024 – September 2026; 208 596 bars

Reading the Bitcoin Clock

- ▣ Weekdays: peak of 16.7 bp at 14:00 UTC, low of 7.3 bp at 10:00 UTC
 - ▶ The peak coincides with the opening of the US equity market and US macro releases
- ▣ Weekend: 5.9 bp against 9.9 bp on weekdays: 1.69 times calmer
- ▣ A market that never closes still follows the clock of banks, ETFs and traditional investors
- ▣ High-frequency Bitcoin data also help forecast daily risk [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)

Liquidity and Volatility: SPY and the VIX



- Daily intraday illiquidity: mean of $|r|$ / traded value over the 78 five-minute bars, in bp per USD 100 million; monthly means

Reading Liquidity and Volatility

- ▣ Correlation of daily illiquidity with the VIX: 0.58; of monthly means: 0.69
- ▣ Regression of log illiquidity on log VIX: elasticity 0.95 (HAC standard error 0.12)
- ▣ April 2025: illiquidity 3.04 against a sample mean of 1.83
 - ▶ Liquidity providers widen quotes and shrink depth exactly when investors most need to trade
- ▣ Liquidity risk is therefore correlated with market risk [Acharya & Pedersen \(2005\)](#)

Recap: Intraday Patterns

- ▣ Volume and volatility are U-shaped over the trading day, with spikes at scheduled news
- ▣ Bitcoin, open 24/7, is most active when US markets are open and calmest at weekends
- ▣ Illiquidity rises with volatility: liquidity disappears when it is most needed

Informed and Uninformed Traders

- A market maker faces two kinds of traders [Bagehot \(1971\)](#)
 - ▶ **Informed**: know something about the value; they buy when it is high and sell when it is low
 - ▶ **Uninformed** (liquidity traders): trade for their own needs, such as savings, redemptions or hedging
- The market maker cannot tell them apart, but knows that a buy order is more likely to come from an informed buyer
- So it sets the ask above the current value estimate and the bid below it: the adverse-selection spread
- Each trade reveals information: prices learn from order flow

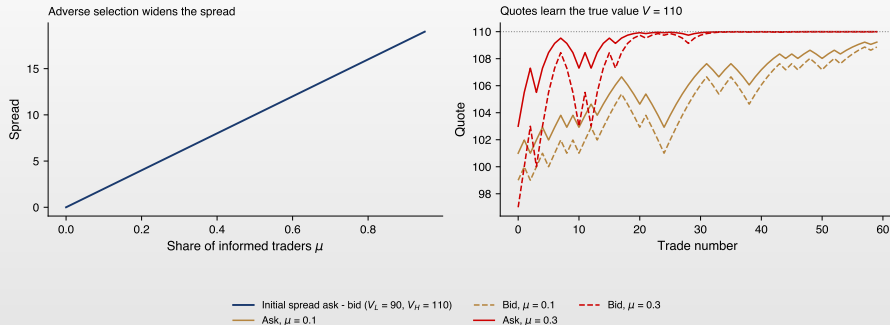
The Glosten–Milgrom Model

- Value $V \in \{V_L, V_H\}$, $P(V = V_H) = \theta$; each period one trader arrives [Glosten & Milgrom \(1985\)](#)
 - ▶ With probability μ informed (buys if $V = V_H$, sells if $V = V_L$), otherwise uninformed (buys or sells with probability $1/2$)
- Competitive, risk-neutral market maker: zero expected profit on each trade
- $a = \mathbb{E}[V \mid \text{buy}]$, $b = \mathbb{E}[V \mid \text{sell}]$
- $P(\text{buy} \mid V_H) = \mu + \frac{1-\mu}{2}$, $P(\text{buy} \mid V_L) = \frac{1-\mu}{2}$
- After each trade, Bayes' rule updates θ : a buy raises it, a sell lowers it

Glosten–Milgrom: the Spread

- $P(V_H \mid \text{buy}) = \frac{\left(\mu + \frac{1-\mu}{2}\right) \theta}{\left(\mu + \frac{1-\mu}{2}\right) \theta + \frac{1-\mu}{2}(1-\theta)}, \quad a = V_L + (V_H - V_L) P(V_H \mid \text{buy})$
- At $\theta = 1/2$: $a - b = \mu(V_H - V_L)$
 - ▶ The spread grows linearly with the share of informed traders
 - ▶ $V_L = 90$, $V_H = 110$: spread 2 for $\mu = 0.1$, 6 for $\mu = 0.3$
- No inventory, no fees: the spread exists purely because of information
- Trade prices form a martingale: the market is efficient in the semi-strong sense (Chapter 2)

Glosten–Milgrom in Action



- Left: initial spread as a function of μ ; right: one simulated sequence of 60 trades with true value $V = 110$ (same random draws for both μ)

Reading the Learning Paths

- With $\mu = 0.3$ the bid exceeds 108 after 7 trades; with $\mu = 0.1$ only after 48
- More informed traders: wider spreads, but faster price discovery
 - ▶ The trade-off between liquidity and informational efficiency
- The spread narrows as the market maker learns: less remains to be learned from the next trade
- Empirical counterpart: the probability of informed trading (PIN) [Easley, Kiefer, O'Hara & Paperman \(1996\)](#)

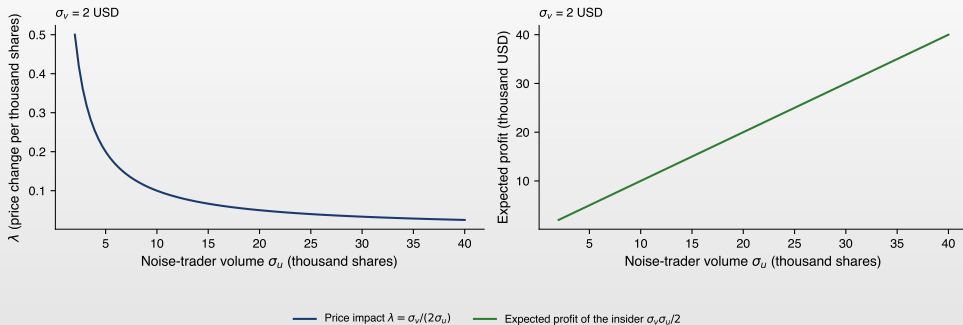
The Kyle Model

- One insider knows $v \sim N(p_0, \sigma_v^2)$ and submits a market order x [Kyle \(1985\)](#)
 - ▶ Noise traders submit $u \sim N(0, \sigma_u^2)$, independent of v
 - ▶ The market maker sees only the total order flow $y = x + u$ and sets $p = \mathbb{E}[v \mid y]$
- The insider chooses x to maximise $\mathbb{E}[(v - p)x]$, knowing that his own order moves the price
- Linear equilibrium: $x = \beta(v - p_0)$, $p = p_0 + \lambda y$

Kyle: the Equilibrium

- $\beta = \frac{\sigma_u}{\sigma_v}, \quad \lambda = \frac{\sigma_v}{2\sigma_u}, \quad \mathbb{E}[\text{profit of the insider}] = \frac{\sigma_v \sigma_u}{2}$
- λ ("Kyle's lambda"): price move per unit of order flow; $1/\lambda$ is **market depth**
 - ▶ More noise trading (σ_u up): deeper market, larger insider trades, larger insider profit
- $\text{Var}(v \mid p) = \sigma_v^2/2$: the price reveals exactly half of the private information
- The insider hides behind noise traders: his order has the same variance as their orders, $\text{Var}(x) = \sigma_u^2$

Kyle's Lambda and the Insider's Profit



□ $\sigma_v = 2$ USD; σ_u from 2 000 to 40 000 shares

From Models to Measurement

- Kyle's λ is estimated by regressing price changes on signed order flow; the Amihud ratio is its daily proxy Hasbrouck (2009)
- PIN: estimated from daily counts of buys and sells, by maximum likelihood Easley, Kiefer, O'Hara & Paperman (1996)
- VPIN: order imbalance measured in volume time, for high-frequency markets Easley, López de Prado & O'Hara (2012)
 - Rises when order flow becomes one-sided, i.e. "toxic" for liquidity providers
- Order-flow imbalance at the best quotes explains most short-horizon price changes Cont, Kukanov & Stoikov (2014)

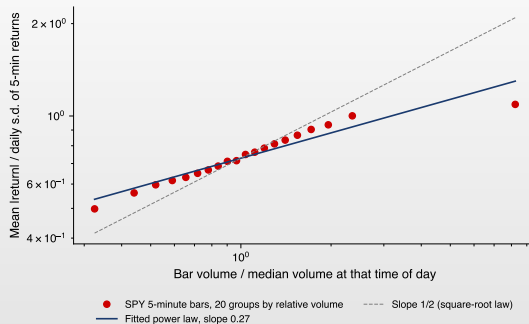
Recap: Information

- ▣ Adverse selection alone creates a spread proportional to the share of informed traders
- ▣ Prices learn from order flow; faster learning comes with wider spreads
- ▣ Kyle's $\lambda = \sigma_v / (2\sigma_u)$: depth grows with noise trading, and half of the information enters the price

Metaorders and the Square-Root Law

- ▣ A **metaorder**: a large order split into many small trades over hours or days
- ▣ Empirical law for the total price move Tóth et al. (2011), Bouchaud, Bonart, Donier & Gould (2018)
 - ▶ $\Delta P \approx Y \sigma_{\text{day}} \sqrt{Q/V_{\text{day}}}$, with Q the metaorder size, V_{day} the daily volume, Y of order 1
- ▣ Concave: trading 1% of daily volume costs about 0.1σ , not 0.01σ
- ▣ Explanation: liquidity near the price is scarce and refills slowly, as in the simulated book of Section 2
- ▣ Testing it needs the traders' own order records; public bars allow only an indirect look (next slide)

Volume and Price Moves in SPY Bars



- 115 052 five-minute bars in 20 equal-size groups by relative volume (bar volume / median volume at that time of day)

Reading the Volume–Move Relation

- ▣ Mean $|r|$, scaled by the day's volatility, rises with relative volume with a slope of 0.27 on log-log axes
- ▣ Concave, as the square-root law predicts, but flatter than $1/2$
- ▣ Not a causal test: a bar mixes buyers and sellers, and news moves both volume and price
 - ▶ Opposite trades cancel, so the net flow is smaller than the volume: this flattens the slope
- ▣ For execution costs use the metaorder law, calibrated on one's own trades

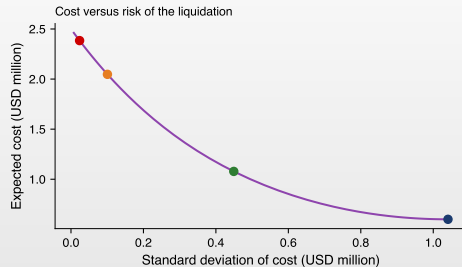
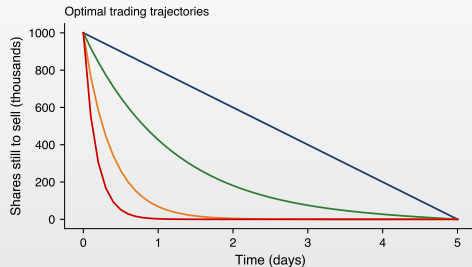
Optimal Execution: the Trade-Off

- Sell X shares within T days, in N steps of length $\tau = T/N$; holdings x_k with $x_0 = X$, $x_N = 0$; trades $n_k = x_{k-1} - x_k$
Almgren & Chriss (2001)
- Price dynamics: $S_k = S_{k-1} + \sigma\sqrt{\tau}\xi_k - \gamma n_k$; execution price $\tilde{S}_k = S_{k-1} - \eta n_k/\tau$
 - ▶ γ : permanent impact; η : temporary impact, paid only by the trade itself
- **Trading fast**: high temporary impact, low price risk; **trading slowly**: the opposite
- Objective: minimise $\mathbb{E}[C] + \lambda \text{Var}[C]$, with C the implementation cost and λ the risk aversion

Almgren–Chriss: the Optimal Trajectory

- $x_k = X \frac{\sinh(\kappa(T - t_k))}{\sinh(\kappa T)}, \quad \frac{2}{\tau^2} (\cosh(\kappa\tau) - 1) = \tilde{\kappa}^2 = \frac{\lambda\sigma^2}{\tilde{\eta}}, \quad \tilde{\eta} = \eta - \frac{\gamma\tau}{2}$
- $\mathbb{E}[C] = \frac{1}{2}\gamma X^2 + \frac{\tilde{\eta}}{\tau} \sum_{k=1}^N n_k^2, \quad \text{Var}[C] = \sigma^2\tau \sum_{k=1}^N x_k^2$
- $\lambda = 0$: $\kappa = 0$ and x_k falls linearly (trade evenly, the time-weighted schedule)
 - Large λ : most of the order is traded at the start, to cut the exposure to price risk
- $1/\kappa$ is the “half-life” of the trade: the time scale over which the position is worked off

Optimal Trajectories and the Efficient Frontier



— Risk-neutral, $\lambda = 0$ (sell evenly) — $\lambda = 2e-5$ — Efficient frontier ($N = 5$ days)
 — $\lambda = 2e-6$ — $\lambda = 1e-4$ (very risk averse)

- Numerical example of [Almgren & Chriss \(2001\)](#): $X = 10^6$ shares, $S_0 = 50$ USD, $T = 5$ days, $\sigma = 0.95$ USD per day^{1/2}, $\eta = 2.5 \cdot 10^{-6}$, $\gamma = 2.5 \cdot 10^{-7}$

Reading the Execution Chart

Strategy (5 daily steps)	Expected cost	Standard deviation
Even selling, $\lambda = 0$	0.60 million USD	1.04 million USD
$\lambda = 2 \cdot 10^{-6}$	1.08	0.45
$\lambda = 2 \cdot 10^{-5}$	2.05	0.10
$\lambda = 10^{-4}$	2.38	0.02

- With $\lambda = 2 \cdot 10^{-6}$ the first day sells 57% of the position; 428 599 shares remain
- Moving from even selling to $\lambda = 2 \cdot 10^{-6}$ cuts the standard deviation from 1.04 to 0.45 million USD, for 0.48 million USD more expected cost
- The frontier is convex: the first units of risk reduction are cheap, the last ones expensive

Execution Benchmarks

- ▣ **Implementation shortfall:** the difference between the paper portfolio at the decision price and the executed one [Perold \(1988\)](#)
- ▣ VWAP (Volume-Weighted Average Price): trade in proportion to the expected volume curve (the U-shape of Section 5)
- ▣ TWAP (Time-Weighted Average Price): equal amounts in each interval, the $\lambda = 0$ solution
- ▣ A participation algorithm (a fixed share of market volume) ignores price
 - ▶ This is the algorithm at the heart of the Flash Crash (next section)

Recap: Impact and Execution

- Impact is concave in size: the square-root law for metaorders
- Optimal execution trades off expected impact cost against price risk
- Risk-averse traders front-load; risk-neutral traders spread the order evenly

High-Frequency Trading



A microwave relay tower (Mojave, California); similar line-of-sight links carry trading signals between Chicago and New Jersey

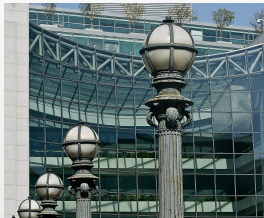
Photo: [Tony Webster \(2016\)](#); CC BY-SA 2.0; [Wikimedia Commons](#)

- HFT (high-frequency trading): computer algorithms that trade in micro- to milliseconds, end the day flat and profit from many tiny trades
- Speed: servers placed next to the exchange's matching engine (co-location), microwave links
 - ▶ Microwaves travel through air faster than light through glass fibre
- Low latency speeds up the incorporation of information and narrows spreads [Hasbrouck & Saar \(2013\)](#)

HFT Firms as the New Market Makers

- A single HFT firm acted as a market maker on Chi-X and Euronext, with most of its trades passive (limit orders) [Menkveld \(2013\)](#)
 - ▶ It earned on the spread and lost on positions: the classic market-making business, at high speed
- High-frequency markets change what “information” means: order flow and latency matter as much as fundamentals [O'Hara \(2015\)](#)
- HFT liquidity is fast to come and fast to leave: plentiful in calm markets, fragile in stress
- Market makers in limit-order markets as intermediaries: [Pele \(2025\)](#)

The Flash Crash of 6 May 2010

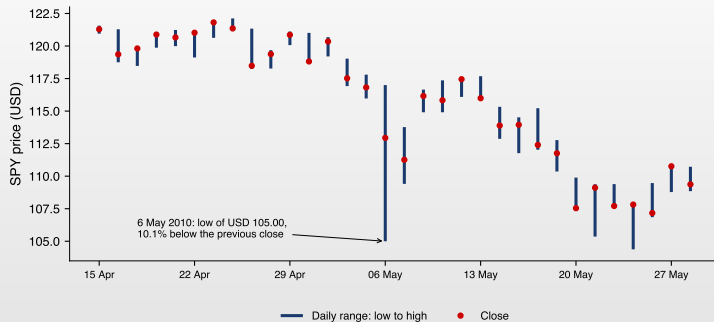


Headquarters of the US Securities and Exchange
Commission, Washington

Photo: David (dbking) (2008); CC BY 2.0; Wikimedia
[Commons](#)

- Report of the US futures (CFTC) and securities (SEC) regulators [CFTC & SEC \(2010\)](#)
- At 2:32 p.m. a mutual fund complex started selling 75 000 E-Mini S&P 500 futures (about USD 4.1 billion)
 - ▶ The algorithm targeted 9% of the previous minute's volume, "without regard to price or time"
- By 2:45:28 p.m. buy-side depth in the E-Mini had fallen below 1% of its morning level
- Over 20 000 trades in more than 300 securities were executed more than 60% away from their earlier prices, some at a penny

SPY Around the Flash Crash

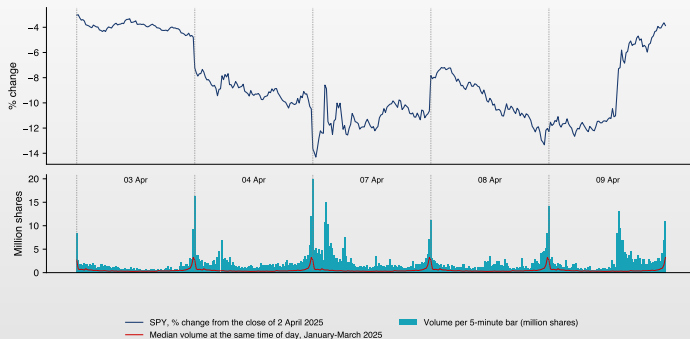


□ Daily low-high range and close, 15 April – 28 May 2010 (prices not adjusted)

Reading the Flash Crash

- ▣ SPY low of USD 105.00: 10.1% below the previous close of 116.82; it still closed only 3.3% lower
- ▣ Daily volume was 3.0 times the median of the previous three weeks
- ▣ HFT firms did not trigger the crash, but their reaction to the selling pressure amplified it [Kirilenko, Kyle, Samadi & Tuzun \(2017\)](#)
 - ▶ A 5-second pause in E-Mini trading at 2:45:28 p.m. let buyers return [CFTC & SEC \(2010\)](#)
- ▣ Responses: single-stock circuit breakers (a 5-minute pause after a 10% move in 5 minutes) and clearer rules for cancelling erroneous trades [CFTC & SEC \(2010\)](#)

A Recent Stress Episode: April 2025



- SPY, 5-minute bars, 3–9 April 2025; bottom panel: bar volume against the median volume at the same time of day in January–March 2025

Reading April 2025

- 7 April: intraday range of 8.6%; one 5-minute bar rose 4.28%, and a bar ten minutes later fell 2.92%
 - ▶ Realised volatility of the day: 6.7%, against a sample median of 0.61%
- 9 April: +10.0% from the open to the close
- Over the five days, volume was 4.6 times its usual level at the same times of day
- Unlike 2010, prices stayed continuous: circuit breakers were not needed, but liquidity was thin (Section 5)

The Speed Race and Frequent Batch Auctions

- In a continuous order book, correlations between related markets break down at millisecond horizons [Budish, Cramton & Shim \(2015\)](#)
 - ▶ This creates arbitrage opportunities that go to the fastest trader: an arms race in speed
- The cost of the race is borne by investors through wider spreads and thinner books
- Proposal: **frequent batch auctions**, e.g. every second, at a single price
- Competition moves from speed to price: narrower spreads and deeper markets in the model

Recap: HFT and Stress

- HFT firms are today's market makers: cheap liquidity in normal times, fragile under stress
- 6 May 2010: a price-insensitive sell algorithm met a book that emptied; prices fell 10% in minutes
- Market design (auctions, pauses, tick sizes) can reduce the value of pure speed

Crypto Exchanges

- ▣ Trading never stops: 24 hours a day, 7 days a week, with no opening or closing auction
- ▣ CEX (Centralised EXchange): a firm runs a limit order book, as on a stock exchange
 - ▶ The same asset trades on dozens of venues at once: liquidity is fragmented
- ▣ DEX (Decentralised EXchange): trades run by smart contracts, often without an order book: an automated market maker (AMM) prices by a formula (Chapter 16)
- ▣ The daily traded values used here sum trading across exchanges

Arbitrage Across Crypto Exchanges

- ▣ The same Bitcoin can trade at different prices on different exchanges [Makarov & Schoar \(2020\)](#)
- ▣ Price gaps are much larger **across countries** than within a country
 - ▶ Capital controls limit the movement of arbitrage capital
- ▣ The gaps open when Bitcoin appreciates strongly: demand shocks hit local markets first
- ▣ A common component of signed volume across exchanges explains 80% of Bitcoin returns [Makarov & Schoar \(2020\)](#)

Measuring Crypto Liquidity

- ▣ High-low estimators capture the **time variation** of crypto liquidity well; Amihud-type ratios capture **levels** and differences between venues [Brauneis, Mestel, Riordan & Theissen \(2021\)](#)
 - ▶ Consistent with our results: spread estimates rank crypto by volatility, Amihud by traded value
- ▣ Liquidity of crypto-assets can be forecast from market data [Agakishiev, Härdle & Pele \(2025\)](#)
- ▣ Bitcoin and Ethereum trade as much value as the largest US stocks; smaller tokens are hundreds of times less liquid
- ▣ Weekend and night hours are calmer and thinner (Section 5)

Recap: Crypto Microstructure

- ▣ Continuous, global and fragmented: many order books for one asset
- ▣ Cross-country price gaps reveal limits to arbitrage
- ▣ Top crypto-assets now rival US large caps in liquidity; the long tail does not

The Bucharest Stock Exchange



Offices of the Bucharest Stock Exchange

Photo: [Za.kurek \(2014\)](#); CC BY-SA 4.0; [Wikimedia](#)

[Commons](#)

- An electronic limit order book, as in the rest of this chapter [BVB \(history\)](#)
- Emerging-market status (FTSE Russell) since September 2020; Banca Transilvania and Nuclearelectrica were the first stocks in FTSE emerging-market indices
- A thin market: herding and bubbles in illiquid markets were studied on the BVB by [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
 - ▶ Bubbles and crashes are the subject of Chapter 17

BVB Liquidity: What the Data Say

- ▣ Amihud ratio of Banca Transilvania: 24.7 bp per USD 1 million; of Transelectrica: 1024.0; of SPY: 0.0016
- ▣ The BET ETF trades less than Banca Transilvania: 212.9 bp per USD 1 million
- ▣ Illiquidity has fallen since 2017, but the BVB remains thousands of times less liquid than US large caps
 - ▶ Daily spread estimates hide this: they rank BVB stocks with US stocks, because volatilities are similar
- ▣ Monthly market returns fall when illiquidity rises unexpectedly, on the BVB as in the US and in crypto (t-statistics -4.0, -3.8, -4.1; seminar Part C)

Recap: The BVB

- ▣ Concentrated trading: two stocks carry more than half of the blue-chip traded value
- ▣ Use traded-value measures (Amihud), not daily spread proxies, to compare the BVB with other markets
- ▣ For portfolio decisions, the illiquidity of the BVB is a cost and a risk (Chapter 4 and seminar Part C)

Summary

- ▣ Prices are formed in limit order books: limit orders supply liquidity, market orders consume it
- ▣ The spread pays for processing, inventory and adverse selection; Glosten–Milgrom and Kyle model the last one
- ▣ Spread proxies from daily data measure volatility; the Amihud ratio separates markets by orders of magnitude
- ▣ Liquidity follows the clock (U-shape) and the VIX; it disappears in stress (2010, 2025)
- ▣ Impact is concave in size; optimal execution balances impact against price risk

Which Liquidity Measure When?

Available data	Suggested measure
Trades and quotes with timestamps	Effective and realised spread, price impact, order-flow imbalance
Intraday bars (1–5 minutes)	Corwin–Schultz or Abdi–Ranaldo per day; intraday Amihud
Daily prices and volumes	Amihud ratio for levels across assets; high-low estimators only for changes over time
Own executions	Implementation shortfall; calibrate the square-root law
Theory, what-if analysis	Simulated order books; Glosten–Milgrom, Kyle, Almgren–Chriss

Coming Next: Continuous-Time Models (Chapter 11)

- ▣ From trades and order books back to smooth models of prices: Brownian motion and Itô calculus
- ▣ At high frequency, microstructure noise contaminates volatility estimates: realised volatility (Chapter 9) needs care
- ▣ Further reading: Hasbrouck (2007), Bouchaud, Bonart, Donier & Gould (2018), Harris (2003), Franke, Härdle & Hafner (2019)

Further Study on Quantinar

Advanced courses on quantinar.com

 Applied Computational Finance

Code and data

 Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_10

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